Name: Date: Period:

Newton #5 (Mixed Net Force Problems)

Remember, all problems must include:

(1) A Picture (Free-body diagram) *(2) Knowns/Unknowns* *(3) Equation with letters*
(4) Plug & Chug *(5) Answer with Units*

$F_{NET} = ma$ $F_{NET} = \pm F_1 \pm F_2 \pm \cdots$ $F_G = mg$ $g = 9.8\,\mathrm{m/s^2}$

1. A skydiver (65 kg) deploys her parachute. How much force does the parachute exert on the skydiver if her acceleration downwards is only 1.2 $\mathrm{m/s^2}$?

2. A giant Snorlax is in your way and your PokeFlute is broken. You need to pull it with a rope. If you apply a force of 520 N and there is 510 N of friction, what is the acceleration of the Snorlax? (According to the PokeDex, the average Snorlax's mass is about 459 kg.)

3. A vanilla coke with a mass of 0.371 kg is sitting at rest on a table. What is the normal force acting on the can?